

UTKARSH SHARMA

✉ utkarsh2024@iisc.ac.in

📄 [Utkarsh Sharma](#)

🌐 [usharma002.github.io](https://github.com/usharma002)

🐙 github.com/USharma002

Education

Indian Institute of Science (IISc)

Bangalore, India

M.Tech in Computer Science and Engineering | **CGPA: 9.1/10.0**

2024 – 2026

Key Coursework: Graphics & Visualization, Learning for 3D Vision and Inverse Graphics, Deep Learning for Computer Vision, Computational Topology, Computational Methods of Optimization, Machine Learning for Signal Processing, Design and Analysis of Algorithms

Chandigarh University

Mohali, India

B.E. in Computer Science | **CGPA: 8.24/10.0**

2020 – 2024

Research Experience

M.Tech Thesis Project: Radiosity-Guided Accelerated Path Tracing | CS 299, IISc

Jan – Apr 2026

- Jointly advised by **Prof. Vijay Natarajan** (Dept. of CSA, IISc) and **Prof. R. Venkatesh Babu** (Dept. of CDS, IISc; Vision & AI Lab).
- Proposed a camera-independent path guiding framework utilizing hardware-accelerated progressive radiosity to extract top- K radiant contributors to construct incoming radiance distribution.
- Designed a closed-form von Mises-Fisher (vMF) mixture model constructed on-the-fly, mapping precomputed radiometric geometry to continuous spherical distributions.
- Implemented a fused CUDA update kernel achieving bounded $\mathcal{O}(T)$ memory scaling and delivering up to $2.4\times$ higher sample throughput than NPM and PPG baselines on Diffuse dominant scenes with comparable variance reduction.
- **Radiosity-Guided Accelerated Path Tracing – Utkarsh Sharma**, Prashant Goswami, Rishubh Parihar, Vijay Natarajan, R. Venkatesh Babu. (*Under review at JCGT, July 2026*)

Work Experience

Netradyne

Bangalore, India

Machine Learning Engineer

July 2026 – Present

- Working within the Data Science team to analyze and refine the **GreenZone® Score**, an AI-driven metric that quantifies driver performance based on large-scale telematics and video-derived data.
- Evaluating correlations between driver behavior indicators (e.g., proactive safe driving, risk events) and safety outcomes to improve predictive risk modeling and fleet optimization algorithms.
- Utilizing Python, SQL, Pandas and NumPy to process and extract actionable insights from millions of miles of real-world driving data.

Selected Research & Projects

Futaba: Hardware-Accelerated GPU Path Tracer | C++17, CUDA, OptiX, OpenGL

Mar – Aug 2025

- Developed a GPU-accelerated path tracer in C++/CUDA with NVIDIA OptiX hardware ray tracing and an optimized software BVH fallback, tuning thread occupancy and GPU memory layouts.
- Implemented Next Event Estimation with multiple importance sampling, Russian Roulette, thin-lens depth of field, environment-map importance sampling, and Beckmann microfacet BSDFs (conductor/dielectric/plastic).
- Built an interactive OpenGL viewport (zero-copy PBO display) with multi-channel G-buffer visualizations (albedo, normals, depth, heatmap) and an OptiX AI denoiser for debugging and final output quality.

Neural Path Guiding for Path Tracing | Mitsuba 3, PyTorch, Tiny-CUDA-NN

Jun – Dec 2025

- Built a modular neural path-guiding framework in Mitsuba 3 (Python), implementing vMF mixture models, Neural Importance Sampling, and Distribution Factorization to guide ray directions, validating $2-4\times$ variance reduction in complex scenes.
- Developed a PyQt-based interactive GUI to inspect Mitsuba renders, visualize spherical probability density functions (PDFs), and debug neural network prediction behavior during rendering.
- Prototyped Neural Radiance Caching using online optimization and spatial hash-grid encodings.

Deep Learning-Based Denoising for Monte Carlo Rendering | PyTorch, PBRT

Jan – Apr 2025

- Evaluated and benchmarked diffusion-based and U-Net denoisers for low-sample Monte Carlo path tracing outputs in PyTorch and PBRT.

- Incorporated G-Buffer features (albedo, normals, depth, motion vectors) as multi-channel inputs to guide predictions, preserving geometric edges and material boundaries.
- Analyzed reconstruction quality using SSIM and PSNR metrics on PBRT-generated test scenes.

Neural Radiance Fields & 3D Gaussian Splatting | *PyTorch*

Jan – May 2025

- Built a NeRF pipeline from scratch in PyTorch, implementing the volumetric rendering equation, sinusoidal positional encoding, and hierarchical stratified sampling to reconstruct 3D scenes.
- Implemented 3D Gaussian Splatting, optimizing 3D covariance matrices and spherical harmonics (SH) coefficients; validated on the NeRF Synthetic dataset (PSNR > 25 dB).

VolPath: Scalar-Field Reconstruction | *Python, PyTorch, HDF5/VTK*

Aug – Dec 2025

- Built a volume-rendering application to reconstruct continuous scalar fields from unstructured scientific point-cloud data, applied to cosmological simulation output from IllustrisTNG (TNG100-1).
- Implemented and compared multiple reconstruction methods — linear, nearest-neighbor, Gaussian RBF, natural-neighbor, octree-based, and learned neural field representations — with an interactive UI for training and checkpointing.

Technical Writing

Introduction to Differentiable Rendering

July 2026

Long-form article on differentiable rendering techniques for graphics and inverse-vision problems.

Introduction to 3D Rendering

Apr 2026

First-principles guide: transforms, cameras, ray/path tracing, SDFs, and volume rendering.

Importance Sampling & Path Guiding for Path Tracing

Dec 2025

Survey of variance-reduction and importance-sampling techniques in Monte Carlo rendering.

3D Reconstruction with 3DGS, NeRF, and Their Variants

Nov 2025

Comparative study of modern 3D reconstruction architectures, strengths, and applications.

Full archive at [usharma002.github.io](https://github.com/usharma002)

Skills & Languages

Programming: C, C++ (17/20), CUDA, Python, GLSL/OpenGL, JavaScript/WebGL

Computer Vision/Graphics: Physically Based Rendering (PBR), GPU Architectures, Ray/Path Tracing, Volume Rendering, Memory Optimization, OpenGL, WebGL

Deep Learning: Vision Transformers, Autoencoders, Normalizing Flows, Diffusion Models, Self-Supervised Learning, Neural Radiance Caching

Frameworks & Debug Tools: PyTorch, Mitsuba 3, PBRT, Tiny-CUDA-NN, Dr.Jit, Three.js, GDB, Git

Spoken Languages: Hindi (Native), English (Fluent)

Teaching Experience

Teaching Assistant – Algorithms & Programming | *Indian Institute of Science*

Aug – Dec 2025

Teaching Assistant – Computational Topology | *Indian Institute of Science*

Jan – May 2026

Academic Achievements

Rank 8 out of 123,967 candidates in GATE Computer Science (CS) 2024

Rank 17 out of 39,210 candidates in GATE Data Science & AI (DA) 2024

Winner, CodeSmash 2.0 Competitive Programming Contest

Finalist, CodeRush 3.0 Competitive Programming Contest